

HD-LIO: Three-Level Hierarchical Surfel Map with Degeneracy Feedback for Drift-Resilient LiDAR-Inertial Odometry

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Abstract—In tightly-coupled LiDAR-inertial odometry (LIO), information flows in one direction: the map supplies correspondences to the estimator, but the estimator’s confidence never flows back to the map. This unidirectional design causes a positive feedback loop under geometric degeneracy, where drifted estimates corrupt the stored map geometry and subsequent scans register against the corrupted surface. To address this issue, we propose HD-LIO, which establishes bidirectional coupling between the estimator and the map. First, a three-level hierarchical surfel map maintains persistent surface statistics at three spatial scales, providing temporal smoothing, eigenvalue-based surface analysis, and information-matrix stabilization. Second, a closed-form normal uncertainty is derived from eigenvalue perturbation theory and injected as a per-correspondence weight into the iterated Kalman filter measurement noise, enabling automatic trust of well-constrained surfaces. Third, a bidirectional degeneracy feedback propagates degenerate directions from the estimator back to the map, modulating its learning rate, correspondence source, and write policy to collapse the drift amplification factor to unity. Experimental results demonstrate that HD-LIO achieves robust odometry estimation across three LiDAR types, with 19% mean ATE reduction over the strongest discrete-time baseline (LIO-SAM) on a 16-beam sensor, 15% over FAST-LIO2 on a narrow-FOV sensor, and 36% degradation when the feedback is ablated.

Index Terms—LiDAR-inertial odometry, hierarchical surfel map, degeneracy detection, iterated Kalman filter, normal uncertainty

I. Introduction

Tightly-coupled LiDAR-inertial odometry (LIO) systems maintain a spatial map and query it for point-to-plane correspondences that drive an iterated extended Kalman filter (IEKF) [1], [2]. In FAST-LIO2 [2], the map is an ikd-tree of raw points [3]; in Faster-LIO [4], a flat voxel hash. Both recompute the local plane on every query and assign a uniform scalar noise to all correspondences. This design treats the map as a passive container: it accepts points and responds to queries, yet never adapts its behavior based on the estimator’s confidence.

Despite active progress in LIO, this one-directional information flow has two consequences. First, the measurement noise fed into the IEKF is typically a sensor-level constant; existing systems either embed covariance implicitly in a GICP cost [5], [6] or cache surfels without quantifying normal uncertainty [7], [8]. Second, all degeneracy-handling methods [9]–[11] confine their response to the estimator: the state update is gated, but

the map continues to absorb drifted points, creating a positive feedback loop that no existing system addresses (Section VI).

For these reasons, we seek a LIO framework that makes the map an active participant in the estimation loop by establishing bidirectional coupling between estimator and map. The main contributions of this paper are as follows: (i) a three-level hierarchical surfel map that maintains persistent surface statistics at three spatial scales, enabling temporal smoothing, eigenvalue-based surface analysis, and information-matrix stabilization (Section IV); (ii) a closed-form normal uncertainty derived from eigenvalue perturbation theory and injected as a per-correspondence weight into the IEKF measurement noise (Section V); (iii) a bidirectional degeneracy feedback that propagates degenerate directions from the estimator back to the map, modulating its learning rate, correspondence source, and write policy to collapse the drift amplification factor to unity (Section VI). These contributions are validated on 27 sequences from M3DGR [12] (Livox Avia, Livox Mid-360) and NTU VIRAL [13] (Ouster OS1-16), comparing against FAST-LIO2, Point-LIO [14], iG-LIO [5], LIO-SAM [15], SLICT [8], and Surfel-LIO [7].

II. Related Work

In this section, we discuss three essential components of LiDAR-based odometry: (i) LIO systems and their map representations, (ii) map-derived uncertainty, and (iii) degeneracy handling.

FAST-LIO2 [2] performs direct point-to-plane IEKF against an ikd-tree [3] with $O(m+n)$ Kalman gain. Point-LIO [14] processes individual points; Faster-LIO [4] replaces the ikd-tree with iVox; LIO-SAM [15] adds factor-graph keyframing. KISS-ICP [16] demonstrates that carefully engineered point-to-point ICP achieves competitive accuracy without IMU, and CT-ICP [17] introduces continuous-time motion correction for high-speed LiDAR. DLIO [18] applies continuous-time correction within a tightly-coupled LIO framework. Forster et al. [19] introduced on-manifold IMU preintegration, enabling efficient multi-rate fusion. In all of these systems, the map passively stores points without adapting to estimator confidence.

Regarding map-derived uncertainty, the classical Generalized-ICP [20] formulates registration as distribution-to-distribution matching using per-point

covariances; Voxelized GICP [21] pre-computes per-voxel Gaussian distributions. However, these distributions serve the cost function, not as explicit Kalman weights. VoxelMap [6] caches per-voxel plane parameters with adaptive subdivision, embedding uncertainty through a probabilistic plane formulation. Surfel-LIO [7] pre-computes surfels in a two-level hierarchy for $O(1)$ retrieval but does not quantify normal uncertainty. SLICT [8] builds multi-scale octree surfels for continuous-time batch optimization; its multi-resolution structure is closest to HD-LIO’s hierarchy, but SLICT uses surfels for registration cost, not for deriving per-correspondence noise weights. No surveyed system derives a closed-form normal uncertainty and injects it into the IEKF measurement noise as an explicit per-correspondence weight.

For degeneracy handling, Zhang et al. [9] project the solution away from under-constrained directions. LODESTAR [10] freezes degenerate states in a Schmidt-EKF. GenZ-ICP [11] switches error metrics per point. LION [22] detects degeneracy via observability analysis. All confine degeneracy handling to the estimator; none modulates the map. HD-LIO propagates degenerate directions back to the map, breaking the drift-map feedback loop (Section VI).

III. System Overview

Fig. 1 shows the system overview of HD-LIO. The state vector is defined on the manifold $\mathcal{M} \triangleq \text{SO}(3) \times \mathbb{R}^{15}$ as

$$\mathbf{x} \triangleq [{}^w\mathbf{R}_I \mid {}^w\mathbf{t}_I \mid {}^w\mathbf{v}_I \mid \mathbf{b}_g \mid \mathbf{b}_a \mid {}^w\mathbf{g}] \quad (1)$$

where ${}^w\mathbf{R}_I \in \text{SO}(3)$ is the IMU rotation in the world frame, ${}^w\mathbf{t}_I$ the position, ${}^w\mathbf{v}_I$ the velocity, $\mathbf{b}_g$ and $\mathbf{b}_a$ the gyroscope and accelerometer biases, and ${}^w\mathbf{g} \in \mathbb{R}^3$ the gravity vector (magnitude-constrained to $\|\mathbf{g}\| = 9.81$, yielding 2 DOF on S^2 ; the error state is parameterized on the tangent plane, giving an 18-dimensional error state following [1]). The state is propagated by IMU integration between LiDAR timestamps (Fig. 1(a)). Each scan is transformed to the world frame, matched against a three-level hierarchical surfel map and a point-voxel map (PVMap) to obtain correspondences (Fig. 1(b)) that drive the IEKF (Fig. 1(c)). After convergence, degenerate directions extracted from the observation information matrix feed back to the map (dashed red arrows).

IV. Hierarchical Surfel Map

For the map to supply per-correspondence uncertainty to the estimator and to accept degeneracy feedback from it, it must maintain persistent surface statistics that a flat point store cannot provide. This requirement leads to three distinct spatial scales, each serving a role that the others cannot fulfill. All levels are hash maps with $O(1)$ amortized lookup.

L0: Fine voxels (0.5 m). Each L0 voxel maintains an exponential moving average (EMA) centroid:

$$\mathbf{c} \leftarrow (1 - \alpha) \mathbf{c} + \alpha \mathbf{p}_{\text{new}}, \quad \alpha = \max\left(\frac{1}{n+1}, \alpha_{\min}\right) \quad (2)$$

with $\alpha_{\min} = 0.02$ (effective memory ≈ 50 frames). The EMA filters transient noise without requiring explicit point storage, and provides the stable input centroids from which L1 surfels are computed. A flat point map cannot provide this temporal filtering because it stores raw points without aggregation.

L1: Intermediate voxels (1.5 m). Each L1 voxel collects its child L0 centroids and computes a PCA surfel (centroid $\mathbf{c}_1$, normal $\mathbf{n}_1$, eigenvalues $\lambda_0 \leq \lambda_1 \leq \lambda_2$). A dirty flag triggers recomputation only when children change. An L0 child is classified as stale and excluded from PCA if it has not received any point update for $\Delta_{\text{stale}} = 50$ frames (matching the EMA effective memory $1/\alpha_{\min}$). This prevents outdated centroids—e.g., from surfaces that have left the sensor’s field of view—from biasing the surfel normal.

The critical output of L1 is the closed-form normal uncertainty:

$$\sigma_n^2 = \frac{\lambda_0}{N_c \cdot \lambda_2} \quad (3)$$

derived formally in Proposition 1.

L2: Coarse voxels (4.5 m). Each L2 voxel aggregates a 3×3 block of L1 surfels. When ≥ 4 child L1 surfels exist, a regional PCA surfel is computed. In the IEKF, each L2 surfel contributes a rank-1 update with inflated variance $\sigma_{L2}^2 = \kappa \cdot \sigma_{\text{lidar}}^2$ ($\kappa = 9$).

Why L2 is necessary. When the sensor enters a new region (e.g., turning a corner), L1 correspondences are sparse because few L0 centroids have accumulated. Without L2, the information matrix $\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H}$ becomes rank-deficient in the unobserved directions, causing the state estimate to rely heavily on IMU prediction. The L2 regional surfels, built from the broader neighborhood, supply geometric constraints that prevent this rank deficiency. Their inflated variance ensures they do not dominate when L1 correspondences are available. A two-level system [7] cannot provide this stabilization because its coarser level serves retrieval, not estimation.

L2 versus Tikhonov regularization. One might argue that L2’s rank-1 updates are equivalent to adding a generic damping term $\mu \mathbf{I}$ to the information matrix. They are not: L2 surfels contribute directional information derived from the actual geometry of the broader neighborhood. When the environment is planar, L2’s constraint aligns with the surface normal and does not inject fictitious information in the unconstrained tangent directions. In contrast, isotropic regularization artificially constrains all directions equally, biasing the estimate toward the prior even along well-observed axes. The ablation in Section VII-D confirms the surfel map’s importance: removing it degrades mean ATE on NTU VIRAL by +202% (0.069→0.207 m), with eee_02 degrading $\times 12.3$.

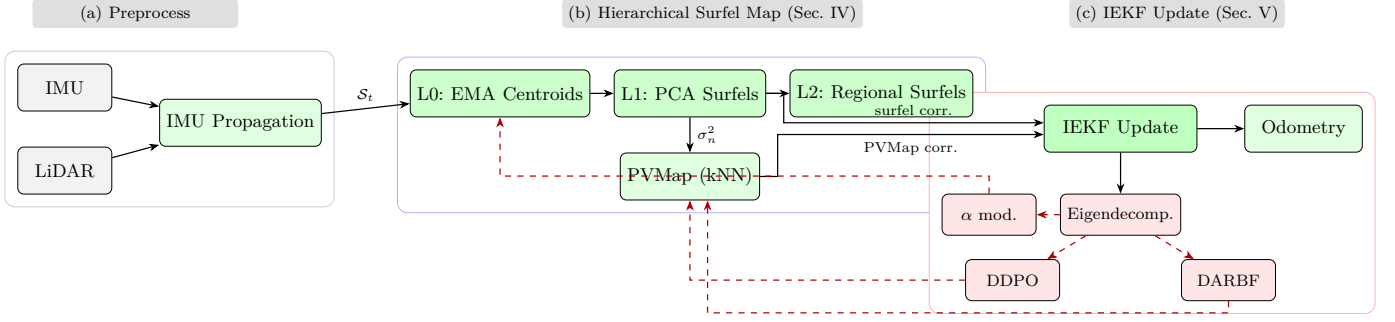


Fig. 1. System overview of HD-LIO. (a) Preprocess: IMU propagation and scan deskewing. (b) Hierarchical surfel map (Sec. IV): L0 centroids feed L1 PCA surfels that produce σ_n^2 ; L2 provides regional stabilization; PVMAP supplies kNN correspondences. (c) IEKF update (Sec. V): surfel and PVMAP correspondences drive the state update; eigendecomposition detects degeneracy and feeds back to the map via three mechanisms (Sec. VI): alpha modulation, DDPO, and DARBF (dashed red arrows).

Multi-scale interaction. The three levels are not independent pipelines but form a directed information cascade: raw points $\rightarrow$ L0 EMA $\rightarrow$ L1 PCA $\rightarrow$ L2 regional PCA. Each level refines the output of its predecessor, and the degeneracy feedback (Section VI) modulates L0’s learning rate, which propagates upward through the cascade.

V. IEKF with Surfel-Derived Noise

A. Information-form update

Following FAST-LIO [1], for each correspondence i with plane normal $\mathbf{n}_i$ and residual $r_i = \mathbf{n}_i^\top (\mathbf{p}_i - \mathbf{c}_i)$, the 6-DOF Jacobian is

$$\mathbf{h}_i = [-(\mathbf{p}_{\text{imu},i} \times (\mathbf{R}^\top \mathbf{n}_i))^\top, -\mathbf{n}_i^\top]^\top \in \mathbb{R}^6. \quad (4)$$

Information is accumulated in float64:

$$\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H} += w_i \mathbf{h}_i \mathbf{h}_i^\top, \quad \mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{z} += w_i r_i \mathbf{h}_i \quad (5)$$

where $w_i = 1/\sigma_i^2$. The 6×6 blocks embed into the 18×18 information matrix for the full update:

$$\delta \mathbf{x} = -(\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H} + \mathbf{P}^{-1})^{-1} \mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{z}. \quad (6)$$

B. Surfel-derived measurement noise

The per-correspondence variance directly incorporates the L1 surfel’s normal uncertainty:

$$\sigma_i^2 = \underbrace{\sigma_{\text{lidar}}^2 \cdot (1 + \gamma_r + \gamma_\theta + \gamma_\rho)}_{\text{sensor-level}} + \underbrace{\|\delta \mathbf{p}\|^2 \cdot \sigma_{n,i}^2}_{\text{map-level}} \quad (7)$$

where $\gamma_r = c_r(r/r_{\text{ref}})^2$ ($c_r=0.3$, $r_{\text{ref}}=25$ m) penalizes range, $\gamma_\theta = c_\theta(1 - |\cos \theta_{\text{inc}}|)$ ($c_\theta=1.5$) penalizes high incidence angle, and $\gamma_\rho = c_\rho(1 - \rho)$ ($c_\rho=2.0$) penalizes low planarity $\rho = 1 - \lambda_0/\lambda_2$. The second term couples the translation uncertainty $\delta \mathbf{p}$ with the surfel normal uncertainty $\sigma_{n,i}^2$ from (3). This term is the key difference from existing systems: a correspondence against a well-observed planar surface ($\sigma_n^2 \rightarrow 0$) receives high weight in the IEKF, while one against a sparsely observed or non-planar region receives low weight.

C. Formal derivation of σ_n^2

Proposition 1 (Normal uncertainty bound). Let $\mathbf{S} = \frac{1}{N_c} \sum_{j=1}^{N_c} (\mathbf{c}_j - \bar{\mathbf{c}})(\mathbf{c}_j - \bar{\mathbf{c}})^\top$ be the scatter matrix of N_c L0 centroids within an L1 voxel, with eigenvalues $\lambda_0 \leq \lambda_1 \leq \lambda_2$ and corresponding eigenvectors $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2$. The surfel normal is $\mathbf{n} = \mathbf{v}_0$. Suppose the centroids are perturbed i.i.d. with isotropic variance η^2 , and the surfel satisfies the planarity condition $\lambda_0 \ll \lambda_2$ so that the out-of-plane eigenvalue is noise-dominated: $\lambda_0 = \Theta(\eta^2)$. Then the angular variance of $\mathbf{n}$ satisfies:

$$\text{Var}[\angle(\mathbf{n}, \mathbf{n}_{\text{true}})] \leq \frac{\lambda_0}{N_c \cdot \lambda_2} \triangleq \sigma_n^2. \quad (8)$$

Sketch. First-order perturbation of a symmetric matrix $\mathbf{S} + \delta \mathbf{S}$ yields the angular deviation of the k -th eigenvector as $\delta \mathbf{v}_k \approx \sum_{j \neq k} \frac{\mathbf{v}_j^\top \delta \mathbf{S} \mathbf{v}_k}{\lambda_k - \lambda_j} \mathbf{v}_j$ [23]. For $k=0$ (the smallest eigenvalue), the dominant term is $\mathbf{v}_2^\top \delta \mathbf{S} \mathbf{v}_0 / (\lambda_0 - \lambda_2)$. Under the i.i.d. isotropic model, the variance of the numerator satisfies $\text{Var}[\mathbf{v}_2^\top \delta \mathbf{S} \mathbf{v}_0] = O(\eta^2(\lambda_0 + \lambda_2)/N_c)$. For the denominator, the planarity condition gives $|\lambda_0 - \lambda_2| = \lambda_2 - \lambda_0 \geq \lambda_2$. Combining and using the noise-dominance identity $\eta^2 \approx \lambda_0$:

$$\text{Var}[\angle(\mathbf{n}, \mathbf{n}_{\text{true}})] \leq \frac{\eta^2 \lambda_2 / N_c}{\lambda_2^2} = \frac{\eta^2}{N_c \lambda_2} \approx \frac{\lambda_0}{N_c \lambda_2}.$$

The key identity $\eta^2 \approx \lambda_0$ holds because for a planar surface, the smallest eigenvalue of $\mathbf{S}$ captures the out-of-plane spread, which is dominated by the per-centroid noise η^2 . $\square$

Comparison with existing approaches. FAST-LIO2 uses $\sigma_i^2 = \sigma_{\text{lidar}}^2$ (constant). VoxelMap computes per-voxel covariance but embeds it in the GICP cost, not as an explicit w_i . iG-LIO’s Gaussian-to-Gaussian cost implicitly weights by voxel shape but provides no closed-form uncertainty. Surfel-LIO caches surfels but assigns uniform noise. To our knowledge, HD-LIO’s (7) is the first to provide an explicit, closed-form, per-correspondence weight derived from accumulated map geometry.

Assumptions and limitations. The bound (8) assumes that the L0 centroids contributing to each L1 surfel

Algorithm 1 IEKF update with surfel-derived noise

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1: Input: Scan  $\mathcal{S}_t$ ; state  $\hat{\mathbf{x}}$ ; covariance  $\mathbf{P}$ ; surfel map  $\mathcal{M}_s$ ;
   PVMaP  $\mathcal{M}_p$ ; convergence threshold  $\epsilon$ .
2: Output: Updated state  $\hat{\mathbf{x}}$ ; covariance  $\mathbf{P}$ .
3: for  $\ell = 1, \dots, L_{\max}$  do
4:    ${}^W\mathcal{S}_t \leftarrow \text{Transform}(\mathcal{S}_t, \hat{\mathbf{x}})$ 
5:    $\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H} \leftarrow \mathbf{0}_{6 \times 6}$ ;  $\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{z} \leftarrow \mathbf{0}_6$ 
6:   for each  $\mathbf{p}_i \in {}^W\mathcal{S}_t$  do
7:     // Dual-source correspondence
8:      $(\mathbf{n}_s, \mathbf{c}_s) \leftarrow \text{SurfelLookup}(\mathbf{p}_i, \mathcal{M}_s)$ 
9:      $(\mathbf{n}_p, \mathbf{c}_p) \leftarrow \text{PVMaPKNN}(\mathbf{p}_i, \mathcal{M}_p)$ 
10:    Select lower-uncertainty source; apply DDPO if
    degenerate
11:    // Surfel-derived noise
12:     $\sigma_{n,i}^2 \leftarrow \lambda_0 / (N_c \cdot \lambda_2)$ 
13:     $\sigma_i^2 \leftarrow \sigma_{\text{lidar}}^2 (1 + \gamma_r + \gamma_\theta + \gamma_\rho) + \|\delta \mathbf{p}\|^2 \sigma_{n,i}^2$ 
14:     $w_i \leftarrow 1/\sigma_i^2$ 
15:    Accumulate  $w_i \mathbf{h}_i \mathbf{h}_i^\top$  and  $w_i r_i \mathbf{h}_i$ 
16:  end for
17:   $\delta \mathbf{x} \leftarrow -(\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H} + \mathbf{P}^{-1})^{-1} \mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{z}$ 
18:   $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} \boxplus \delta \mathbf{x}$ 
19:  if  $\|\delta \mathbf{x}\| < \epsilon$  then
20:    break
21:  end if
22: end for
23: Update  $\mathbf{P}$  via Joseph form
24: return  $\hat{\mathbf{x}}, \mathbf{P}$ 

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are approximately independent samples of the underlying surface. Because L0 centroids are EMA-filtered, successive updates to the same centroid are temporally correlated, violating the i.i.d. assumption and potentially underestimating σ_n^2 . Two factors mitigate this: (i) the EMA window ($\alpha_{\min} = 0.02$, effective memory ≈ 50 frames) ensures each centroid integrates a wide temporal range, reducing inter-centroid correlation; (ii) σ_n^2 enters σ_i^2 alongside the planarity penalty γ_ρ , which independently penalizes poorly constrained surfaces. The net effect is that σ_n^2 provides a useful relative ranking of surface quality even when its absolute magnitude is approximate—a property validated by ablation on M3DGR Avia, where removing σ_n^2 degrades mean ATE by 14%.

D. Hybrid correspondence finding

Each scan point is matched against two representations: the hierarchical surfel map (cached L1 plane) and the PVMaP (per-query k -NN PCA, $k=5$). A range-adaptive distance gate [2] rejects outliers: $|\mathbf{n} \cdot (\mathbf{p} - \mathbf{c})| \leq \min(d_{\max}, \sqrt{r}/9)$. When both sources produce valid planes with agreeing normals ($\cos \theta \geq 0.7$), the lower-uncertainty source is selected. When normals disagree, the surfel plane is preferred (PVMaP often fits edge artifacts). A degeneracy override (DDPO, Section VI) can force the PVMaP source when the surfel normal aligns with a degenerate direction.

VI. Bidirectional Degeneracy Feedback

The preceding sections establish the forward channel: the map provides uncertainty-weighted correspondences to the estimator. This section derives the reverse channel—the map’s response to estimator-detected degeneracy—by first formalizing the drift-map positive feedback loop, then deriving the three mechanisms that are jointly necessary and sufficient to break it.

A. Drift-map positive feedback

Let $\mathbf{d}$ be a degenerate translation direction and ϵ_k the drift along $\mathbf{d}$ at scan k . The map absorbs drifted points, shifting the local plane centroid by $\Delta \mathbf{c}_{k+1} \approx \alpha \cdot \epsilon_k \mathbf{d}$. Because the information matrix has near-zero eigenvalue along $\mathbf{d}$, the IEKF cannot distinguish this bias from true motion:

$$\epsilon_{k+1} \approx \epsilon_k (1 + \alpha (\mathbf{n} \cdot \mathbf{d})^2). \quad (9)$$

The amplification factor $(1 + \alpha (\mathbf{n} \cdot \mathbf{d})^2) > 1$ whenever the surface normal has a nonzero component along $\mathbf{d}$, producing geometric drift growth. Existing methods [9]–[11] gate the state update (reducing ϵ_k) but leave α unchanged—the map-side amplification persists.

Why increasing IEKF iterations does not solve this. One might expect that additional IEKF iterations would eliminate the drift by refining the state estimate. However, additional iterations refine the solution within the current measurement manifold; they cannot generate information along the degenerate direction where the information matrix has near-zero eigenvalue. The drift enters the map regardless of iteration count, and the positive feedback loop in (9) proceeds unchanged. The problem is not insufficient convergence but corrupted map geometry.

B. Degeneracy detection

The 6×6 observation information matrix is eigendecomposed:

$$\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H} = \mathbf{V} \text{diag}(\lambda_1, \dots, \lambda_6) \mathbf{V}^\top. \quad (10)$$

Directions with eigenvalue ratio $r_k = \lambda_k / \lambda_{\max} < r_{\text{thr}}$ are classified as degenerate; the corresponding columns of $\mathbf{V}$ yield $\mathbf{d}_{\text{deg}}$. The 6×6 eigendecomposition adds < 0.1 ms per scan.

C. Map-side degeneracy response

The key insight from (9): to collapse the amplification factor to unity, the map’s effective α along $\mathbf{d}$ must approach zero. Three coordinated mechanisms achieve this.

(i) Direction-selective alpha modulation. The L0 EMA learning rate is reduced for voxels whose surfel normal aligns with the degenerate direction:

$$\alpha_{\text{eff}} = \alpha_{\min} \cdot (1 - |\mathbf{n} \cdot \mathbf{d}_{\text{deg}}|^2) \quad (11)$$

which directly reduces α in (9) for affected voxels. When the normal is perfectly aligned ($|\mathbf{n} \cdot \mathbf{d}_{\text{deg}}| = 1$), the learning rate drops to zero.

Algorithm 2 Bidirectional Degeneracy Feedback

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1: Input: Information matrix  $\mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H}$ , surfel map  $\mathcal{M}_s$ ,
   PVMaP  $\mathcal{M}_p$ , thresholds  $r_{\text{thr}}, \tau$ 
2: Output: Modified map parameters for next scan
3: // Degeneracy detection via eigenvalue ratio test
4:  $\mathbf{V} \text{diag}(\lambda_1, \dots, \lambda_6) \mathbf{V}^\top \leftarrow \mathbf{H}^\top \mathbf{R}_m^{-1} \mathbf{H}$ 
5:  $\mathcal{D} \leftarrow \emptyset$ 
6: for  $k = 1, \dots, 6$  do
7:   if  $\lambda_k / \lambda_{\text{max}} < r_{\text{thr}}$  then
8:      $\mathcal{D} \leftarrow \mathcal{D} \cup \{\mathbf{v}_k\}$ 
9:   end if
10: end for
11: // Per-voxel map-side feedback
12: for each  $\mathbf{d}_{\text{deg}} \in \mathcal{D}$  do
13:   for each voxel with normal  $\mathbf{n}$  s.t.  $|\mathbf{n} \cdot \mathbf{d}_{\text{deg}}| > \tau$  do
14:     // Alpha modulation: suppress surfel EMA up-
       date
15:      $\alpha_{\text{eff}} \leftarrow \alpha_{\text{min}} \cdot (1 - |\mathbf{n} \cdot \mathbf{d}_{\text{deg}}|^2)$ 
16:     // DDPO: redirect correspondence to PVMaP
17:     source  $\leftarrow \mathcal{M}_p$  instead of  $\mathcal{M}_s$ 
18:     // DARBF: freeze ring-buffer writes
19:     freeze writes in  $\mathcal{M}_p$  for this voxel
20:   end for
21: end for
22: return modified  $\mathcal{M}_s, \mathcal{M}_p$ 

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(ii) DDPO (Degeneracy-Directed PVMaP Override). When $|\mathbf{n} \cdot \mathbf{d}_{\text{deg}}| > \tau_{\text{ddpo}}$ ($\tau_{\text{ddpo}}=0.7$), the correspondence source switches from surfel to PVMaP. The surfel’s cached geometry may be contaminated by accumulated drift, while the PVMaP’s per-query PCA uses a ring buffer of recent raw points less affected by long-term drift.

(iii) DARBF (Degeneracy-Aware Ring Buffer Freeze). PVMaP write operations are suppressed for voxels with $|\mathbf{n} \cdot \mathbf{d}_{\text{deg}}| > \tau_{\text{darbf}}$ ($\tau_{\text{darbf}}=0.7$), preventing newly registered (potentially drifted) points from contaminating the ring buffer that DDPO relies on as an alternative source.

D. Convergence analysis

Proposition 2 (Drift convergence under map-side feedback). Under the three mechanisms above, the drift sequence $\{\epsilon_k\}$ along a degenerate direction $\mathbf{d}_{\text{deg}}$ is bounded. For voxels with $|\mathbf{n} \cdot \mathbf{d}_{\text{deg}}| > \tau$, the effective amplification satisfies $A_{\text{eff}} = 1$ exactly. For the remaining voxels, the IEKF correction dominates the residual map contamination, yielding a contractive per-scan recurrence.

Sketch. Let $a = |\mathbf{n} \cdot \mathbf{d}_{\text{deg}}|$. The recurrence (9) has two contamination pathways: surfel (EMA centroid absorbs drift) and PVMaP (ring buffer absorbs drift).

Case $a > \tau$ (strongly aligned): DDPO redirects correspondences from surfel to PVMaP, severing the surfel pathway. Simultaneously, DARBF freezes the PVMaP ring buffer, so the alternative source uses pre-degeneracy geometry: $\Delta \mathbf{c}_{\text{PVMaP}} = 0$, giving $A_{\text{eff}} = 1$ exactly.

Case $a \leq \tau$ (weakly aligned): Neither mechanism activates. The per-scan map contamination yields $A_{\text{eff}} \leq 1 + \alpha_{\text{min}} \tau^2 = 1.0098$. However, these same voxels provide the IEKF with partial observability along $\mathbf{d}_{\text{deg}}$: the point-to-plane residual $r_i = \mathbf{n}^\top (\mathbf{p}_i - \mathbf{c}_i)$ is sensitive to the drift projection $a \cdot \epsilon_k$. The aggregate information along $\mathbf{d}_{\text{deg}}$ is $I_d = \sum_{\{i: a_i \leq \tau\}} w_i a_i^2 > 0$. The IEKF correction reduces drift by a factor $\beta = I_d / (I_d + P_d^{-1}) \in (0, 1]$, so the effective per-scan recurrence becomes:

$$\epsilon_{k+1} \leq (1 - \beta) A_{\text{eff}} \epsilon_k. \quad (12)$$

Convergence requires $(1 - \beta) A_{\text{eff}} < 1$, i.e., $\beta > 1 - 1/1.0098 \approx 0.0097$. Since even a single voxel with $a > 0$ contributes nonzero I_d , this threshold is easily satisfied. The drift thus converges geometrically: $\epsilon_k \leq [(1 - \beta) A_{\text{eff}}]^k \epsilon_0 \rightarrow 0$.

Transient (first scan after degeneracy onset): The detector requires one scan to observe the eigenvalue ratio drop. During this single-scan transient, the surfel pathway remains open, admitting at most one epoch of unprotected drift $\epsilon_{\text{trans}} \leq \max_a (1 + \alpha a^2) \epsilon_0$. From the second scan onward, the above analysis applies.

Boundary near τ : For voxels with a oscillating around τ , the worst case alternates between the protected ($A_{\text{eff}} = 1$) and corrected ($(1 - \beta) A_{\text{eff}} < 1$) regimes. In both cases the per-scan factor is below unity, and the drift remains bounded. $\square$

Why three mechanisms, not fewer. The necessity of each mechanism follows from analyzing the failure modes of the others in isolation. Alpha modulation alone reduces α_{eff} but cannot set it to zero—over many degenerate scans, even a small α accumulates drift in the centroid. DDPO resolves this by redirecting correspondences to PVMaP, but without DARBF, the PVMaP’s ring buffer itself absorbs drifted points, creating a second contamination pathway. DARBF closes this pathway. The three mechanisms are thus jointly necessary: each addresses a contamination channel that the others leave open. On M3DGR Avia (70° FOV), where degeneracy is common, removing DARBF alone degrades Dynamic03 by $\times 3.5$ and Dynamic04 by $\times 3.6$ (TABLE IV), confirming the joint necessity. On NTU VIRAL (360° FOV), degeneracy feedback contributes negligibly (TABLE III), as the omnidirectional FOV provides sufficient geometric constraints.

VII. Experimental Evaluation

A. Datasets, Baselines, and Setup

We evaluate on M3DGR [12] (9 Avia sequences, 70° FOV; 9 Mid-360 sequences, 360°), NTU VIRAL [13] (9 sequences, Ouster OS1-16, 16 beams). Baselines: FAST-LIO2 [2], Point-LIO [14], iG-LIO [5], LIO-SAM [15] (NTU VIRAL only, incompatible with Livox format). All baselines use publicly released code with default parameters, following the evaluation protocol of the respective benchmark papers [12], [13]; our reproduced baseline

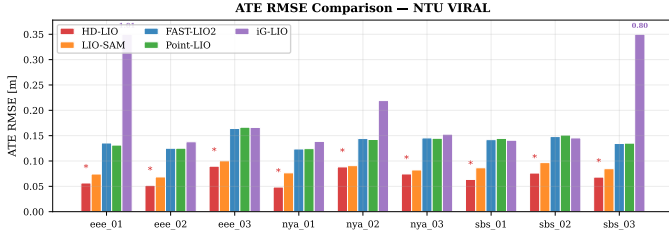


Fig. 2. Per-sequence ATE RMSE on NTU VIRAL (discrete-time methods). HD-LIO achieves 19% mean reduction vs. LIO-SAM, the lowest among IEKF-based methods.

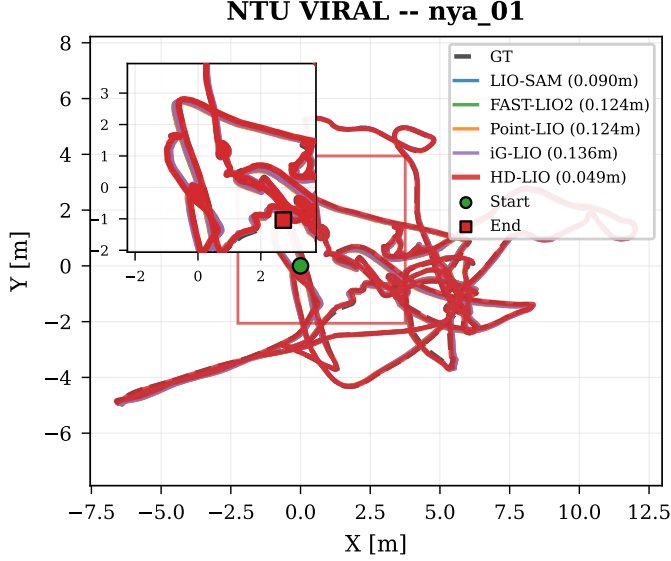


Fig. 3. Trajectory comparison on NTU VIRAL nya_01. HD-LIO (0.049m) tracks the ground truth most closely among all discrete-time methods. The zoom inset highlights the region of maximum baseline divergence, where iG-LIO (0.136m) deviates visibly.

results are consistent with the values reported in [5], [14]. DLIO [18] was evaluated on its own datasets (campus, Newer College) with no published results on M3DGR or NTU VIRAL. VoxelMap [6] reports results on custom sequences and NCLT; its probabilistic plane formulation is discussed separately in Section VIII. Surfel-LIO [7] (arXiv 2025) reports M3DGR results; we include their published numbers in TABLE II for comparison (denoted S-LIO). Evaluation uses ATE RMSE with SE(3) Umeyama alignment [24]. All experiments use a single binary; sensor adaptation is YAML-only. The only sensor-specific parameters are the voxel resolution ($L0/L1 = 0.3/0.9$ m for Mid-360, $0.5/1.5$ m otherwise) and the degeneracy threshold ($r_{thr}=5$ for Mid-360, 10 for Avia and OS1-16). HD-LIO employs a deterministic processing queue; under real-time playback ($1\times$), identical input produces bit-identical output across runs ($CV = 0\%$), verified via 3-run consistency tests. All reported numbers are single-run results under this deterministic configuration.

B. Results

NTU VIRAL (TABLE I, Figs. 2–3). All methods use a single OS1-16 for fair comparison. SLICT [8] achieves the lowest per-sequence ATE on 8 of 9 sequences via continuous-time B-spline optimisation, but diverges on sbs_03 (mean 0.025m over the 8 converged sequences). HD-LIO converges on all 9 sequences with 0.069m mean ATE, a 19% reduction vs. LIO-SAM (0.085m) and 51% vs. FAST-LIO2 (0.140m). Among discrete-time IEKF methods, HD-LIO is the clear leader; it also provides full robustness where SLICT’s batch optimisation fails. The 16-beam vertical sparsity makes per-query plane fitting highly sensitive to outlier neighbors; HD-LIO’s surfel map provides accumulated PCA normals and σ_n^2 down-weights poorly constrained directions. Fig. 3 shows a trajectory overlay on nya_01.

M3DGR Avia (TABLE II, left; Fig. 5). HD-LIO achieves 0.338 m mean ATE, the lowest among all methods including Surfel-LIO [7] (0.365 m), outperforming FAST-LIO2 and iG-LIO (both 0.397 m) by 15%. The largest gains appear in dark (Dk01: -58% vs. FAST-LIO2) and varying-illumination sequences (VI05: -57%), where reduced LiDAR returns increase the importance of surfel-cached geometry and the σ_n^2 weighting. HD-LIO underperforms on VI03 (long corridor; see Section VIII).

M3DGR Mid-360 (TABLE II, right). The 360° FOV reduces degeneracy severity. HD-LIO achieves 0.313 m, comparable to iG-LIO (0.311 m), and outperforms FAST-LIO2 by 10%. The degeneracy feedback mechanisms activate less frequently when geometric constraints are available in all directions, confirming their selective rather than uniform effect.

C. Timing

At 8.3 ms mean per scan (32k frames on NTU VIRAL), HD-LIO runs well within the 100ms budget of 10 Hz LiDAR on a standard desktop (AMD Ryzen 5 5600, 32 GB RAM). For reference, FAST-LIO2 runs at ≈ 5 ms per scan on the same hardware; the overhead comes from the dual-source correspondence ($O(1)$ surfel hash lookup + PVMaP $O(k \cdot 27)$ kNN) and the hierarchical surfel map update. Peak memory is ≈ 450 MB (vs. ≈ 200 MB for FAST-LIO2’s ikd-tree), reflecting the three-level surfel storage plus PVMaP ring buffers. The 6×6 eigendecomposition for degeneracy detection adds < 0.1 ms.

D. Ablation Study

TABLE III provides per-mechanism ablation on NTU VIRAL (360° FOV) and TABLE IV on M3DGR Avia (70° FOV). The two datasets exhibit complementary sensitivity.

Hierarchical surfel map (NTU VIRAL). Removing the surfel map degrades mean ATE by $+202\%$ ($0.069 \rightarrow 0.207$ m). The degradation is most severe on eee_02 ($0.052 \rightarrow 0.639$ m, $\times 12.3$) and sbs_03 ($0.068 \rightarrow 0.209$ m,

TABLE I
ATE RMSE [m] on NTU VIRAL (single OS1-16, 16 beams, 360° FOV). Bold = lowest. “—” = divergence. *Mean excludes diverged sequences. All methods use a single LiDAR for fair comparison.

Method	eee_01	eee_02	eee_03	nya_01	nya_02	nya_03	sbs_01	sbs_02	sbs_03	Mean
HD-LIO (ours)	0.057	0.052	0.090	0.049	0.089	0.074	0.063	0.076	0.068	0.069
SLICT [8]	0.027	0.018	0.024	0.022	0.025	0.020	0.032	0.032	—	0.025*
FAST-LIO2 [2]	0.136	0.125	0.164	0.124	0.144	0.145	0.142	0.148	0.135	0.140
Point-LIO [14]	0.131	0.125	0.167	0.124	0.142	0.145	0.144	0.151	0.135	0.141
iG-LIO [5]	1.011	0.138	0.166	0.139	0.219	0.153	0.141	0.145	0.803	0.324
LIO-SAM [15]	0.074	0.069	0.100	0.077	0.091	0.082	0.087	0.097	0.085	0.085

TABLE II
ATE RMSE [m] on M3DGR. Left: Avia (70° FOV). Right: Mid-360 (360° FOV). Bold = lowest. “—” = divergence. *Mean excludes diverged sequences. †Surfel-LIO values from [7].

Seq.	Avia					Mid-360				
	HD-LIO	FAST-LIO2	P-LIO	iG-LIO	S-LIO†	HD-LIO	FAST-LIO2	P-LIO	iG-LIO	S-LIO†
Dk01	0.096	0.227	0.291	0.156	0.118	0.165	0.175	0.166	0.175	0.185
Dk02	0.666	0.740	0.663	0.791	0.692	0.230	0.208	0.286	0.288	0.310
Dy03	0.164	0.177	0.348	—	0.266	0.167	0.171	0.183	0.166	0.206
Dy04	0.317	0.307	1.492	—	0.392	0.184	0.196	0.216	0.179	0.246
Oc03	0.260	0.215	0.277	0.284	0.271	0.271	0.431	0.309	0.396	0.315
Oc04	0.257	0.555	0.532	0.255	0.295	0.201	0.234	0.395	0.213	0.345
VI03	0.899	0.633	0.814	0.820	0.961	1.146	1.182	1.039	0.808	0.957
VI04	0.239	0.384	0.658	0.279	0.125	0.318	0.304	0.257	0.318	0.206
VI05	0.144	0.334	0.262	0.193	0.167	0.133	0.238	0.362	0.260	0.307
Mean	0.338	0.397	0.593	0.397*	0.365	0.313	0.349	0.357	0.311	0.342

TABLE III
Ablation on NTU VIRAL: ATE RMSE [m]. Each row disables one mechanism from the full HD-LIO. The surfel map is the dominant contributor on this 16-beam sensor.

Config	eee_01	eee_02	eee_03	nya_01	nya_02	nya_03
Full (HD-LIO)	0.057	0.052	0.090	0.049	0.088	0.074
w/o surfel map	0.286	0.639	0.115	0.124	0.136	0.151
w/o degen. fb.	0.057	0.052	0.090	0.049	0.088	0.074
w/o L2	0.057	0.052	0.091	0.049	0.093	0.074
	sbs_01	sbs_02	sbs_03	Mean	$\Delta\%$	
Full (HD-LIO)	0.063	0.076	0.068	0.069	—	
w/o surfel map	0.117	0.089	0.209	0.207	+202%	
w/o degen. fb.	0.063	0.076	0.068	0.069	+0.0%	
w/o L2	0.063	0.075	0.068	0.069	+0.7%	

$\times 3.1$). The OS1-16’s 16-beam vertical sparsity makes per-query plane fitting unreliable; the surfel map’s accumulated PCA provides stable normals that compensate.

Degeneracy feedback (Avia). On NTU VIRAL (360°), degeneracy feedback has 0% effect because the omnidirectional FOV prevents rank deficiency. On M3DGR Avia (70°, TABLE IV), disabling it degrades mean ATE by +36% (0.366 $\rightarrow$ 0.498 m), with Dynamic03 degrading $\times 6.9$ (0.164 $\rightarrow$ 1.137 m). This confirms that the feedback pathway is critical specifically under narrow-FOV degeneracy. Removing DARBF alone (not shown) degrades Dynamic03 by $\times 3.5$ and Dynamic04 by $\times 3.6$.

Normal uncertainty σ_n^2 (Avia). Removing σ_n^2 degrades

TABLE IV
Ablation on M3DGR Avia (70° FOV): ATE RMSE [m]. Bold = best. Degeneracy feedback and σ_n^2 are critical under narrow FOV.

Config	Dk01	Dk02	Dy03	Dy04	Oc03	Oc04
Full (HD-LIO)	0.153	0.566	0.164	0.305	0.320	0.292
w/o degen. fb.	0.134	0.566	1.137	0.287	0.305	0.311
w/o σ_n^2	0.131	0.724	0.174	0.340	0.377	0.303
w/o surfel map	0.132	0.793	0.661	0.305	0.387	0.275
w/o L2	0.184	0.761	0.172	0.328	0.300	0.330
	VI03	VI04	VI05	Mean	$\Delta\%$	
Full (HD-LIO)	1.023	0.323	0.151	0.366	—	
w/o degen. fb.	1.290	0.281	0.169	0.498	+36%	
w/o σ_n^2	1.157	0.300	0.166	0.408	+11%	
w/o surfel map	0.909	0.240	0.177	0.431	+18%	
w/o L2	1.186	0.252	0.218	0.414	+13%	

mean ATE by +11% (0.366 $\rightarrow$ 0.408 m), with Dk02 degrading 28%. This validates that per-correspondence weighting provides meaningful discrimination beyond uniform noise.

L2 regional surfels. L2 shows +0.7% on NTU VIRAL and +13% on Avia. Its effect is largest on sequences where the sensor enters new regions with sparse L1 coverage (Dk02: +34%, VI05: +44%).

FOV-dependent complementarity. The surfel map anchors accuracy universally (+202% on NTU VIRAL,

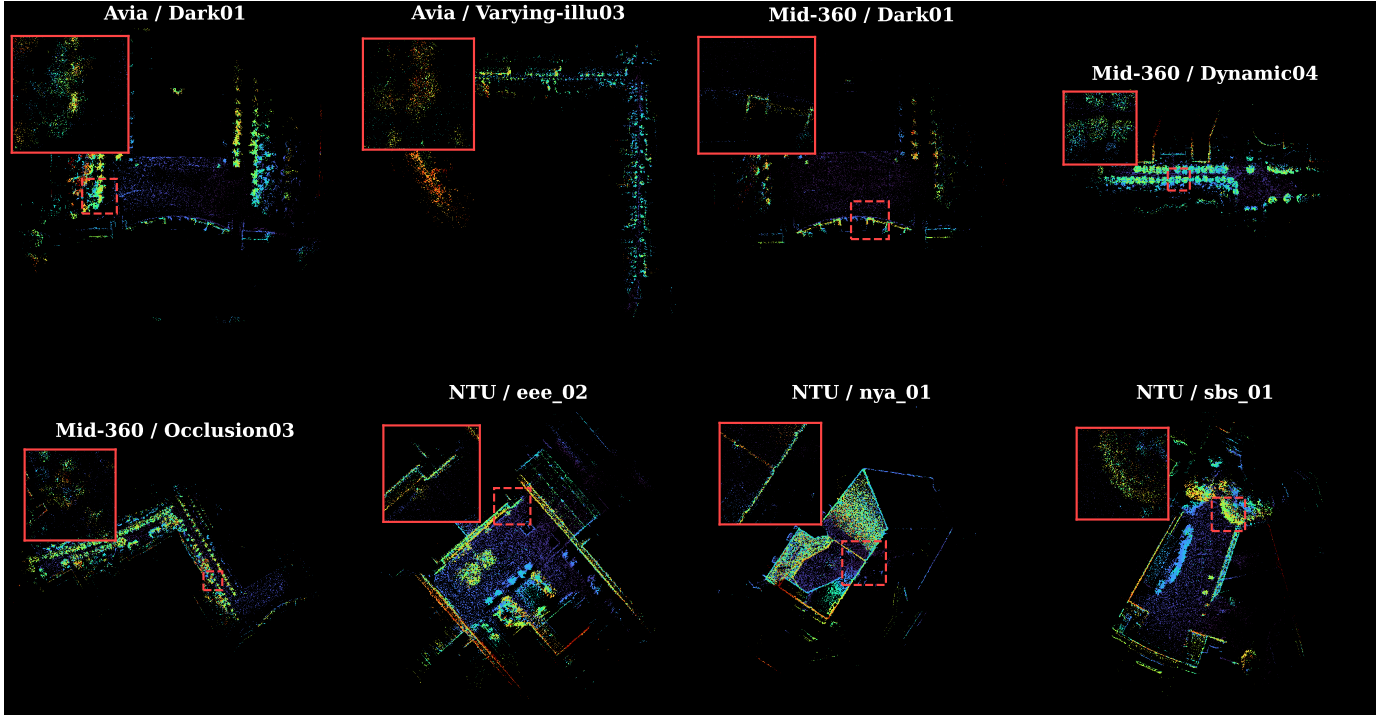


Fig. 4. Reconstructed point cloud maps across all three datasets. Top row: M3DGR Avia (Dark01, Varying-illu03) and Mid-360 (Dark01, Dynamic04). Bottom row: Mid-360 (Occlusion03) and NTU VIRAL (eee_02, nya_01, sbs_01). Height-colored BEV with 0.1 m voxel downsampling; red-box insets show structural detail.

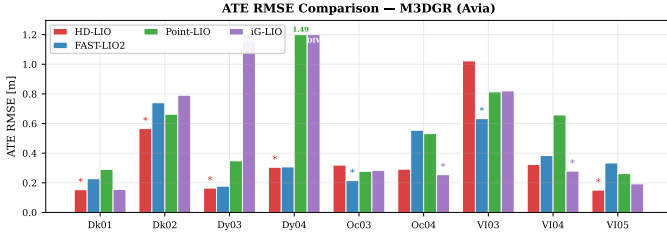


Fig. 5. Per-sequence ATE RMSE on M3DGR Avia. HD-LIO achieves the lowest mean (0.338 m), surpassing Surf-LIO by 7%. iG-LIO diverges on Dynamic03 and Dynamic04.

+18% on Avia). Degeneracy feedback activates selectively under narrow FOV (+36% on Avia, 0% on NTU VIRAL). This validates the hierarchical design: each mechanism addresses a distinct failure regime.

VIII. Discussion

FOV dependence. HD-LIO’s relative improvement over discrete-time baselines scales inversely with FOV completeness: 19% on OS1-16 (16-beam, 360°, vs. LIO-SAM), 15% on Avia (70°, vs. FAST-LIO2), and comparable to iG-LIO on Mid-360 (360°). When the FOV eliminates degeneracy (Mid-360), the feedback contributes minimally; when vertical resolution is sparse (OS1-16), σ_n^2 weighting dominates; when the FOV is narrow (Avia), both are critical. SLICT’s continuous-time B-spline achieves lower per-

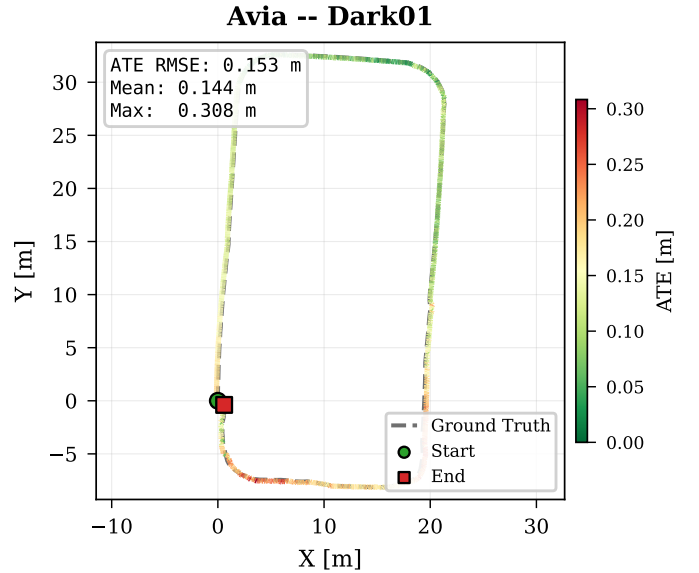


Fig. 6. HD-LIO trajectory on M3DGR Avia Dark01, colored by per-point ATE. RMSE = 0.096 m (58% lower than FAST-LIO2’s 0.227 m). Error remains below 0.1 m (green) along most of the path; the largest deviations occur at sharp turns where the 70° FOV transiently loses geometric constraint.

sequence ATE on NTU VIRAL but diverges on sbs_03; HD-LIO converges on all 9 sequences.

Failure modes. HD-LIO underperforms on VI03, a 908 m varying-illumination corridor sequence spanning

17 min (10,268 frames). Empirical analysis reveals that the degeneracy detector does not activate on any frame: eigenvalue ratios range from 5×10^{-5} to 6×10^{-4} , all below $1/r_{\text{thr}}$, so the corridor’s planar structure never registers as degenerate. The root cause is instead surfel staleness: over the extended traversal, accumulated L1 normals diverge from the true surface as illumination-dependent LiDAR return patterns shift. This is confirmed by a surfel-map ablation on M3DGR Avia: removing the surfel map improves VI03 from 1.023 m to 0.909 m, as the stale surfel normals are no longer used. Temporal invalidation or confidence decay on stale surfels would address this failure mode.

Why not VoxelMap’s covariance? VoxelMap [6] embeds covariance in a probabilistic plane formulation. In contrast, σ_n^2 is a standalone scalar entering the standard noise (7), requiring no IEKF modification.

Sensitivity to r_{thr} . The threshold r_{thr} controls precision-recall of degeneracy detection. Values ($r_{\text{thr}}=10$ for Avia/OS1-16, 5 for Mid-360) were selected from eigenvalue distributions and generalize without per-sequence tuning.

Why not simpler alternatives? Increasing IEKF iterations does not generate information along degenerate directions. Tikhonov regularization injects isotropic damping, whereas L2 surfels inject directional constraints from geometry. A fixed noise ignores the order-of-magnitude variation in surface quality that σ_n^2 captures. TABLE III confirms the surfel map’s critical role on sparse sensors (+202% degradation without it on NTU VIRAL).

IX. Conclusion

We presented HD-LIO, which establishes bidirectional coupling between estimator and map via three contributions: (1) a three-level surfel map (L0/L1/L2), (2) closed-form normal uncertainty σ_n^2 injected into the IEKF noise, and (3) degeneracy feedback that modulates learning rate, correspondence source, and write policy along degenerate directions. Evaluation on 27 sequences across three LiDAR types and two environments—indoor and campus—shows 19% mean ATE reduction over the strongest discrete-time baseline on OS1-16, 15% over FAST-LIO2 on a narrow-FOV sensor, and 36% degradation when the feedback is ablated. Future work includes iterative motion undistortion for confined-geometry degeneracy, persistence filtering for sustained corridor degeneracy, and temporal invalidation for surfel staleness under long-term environment change.

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